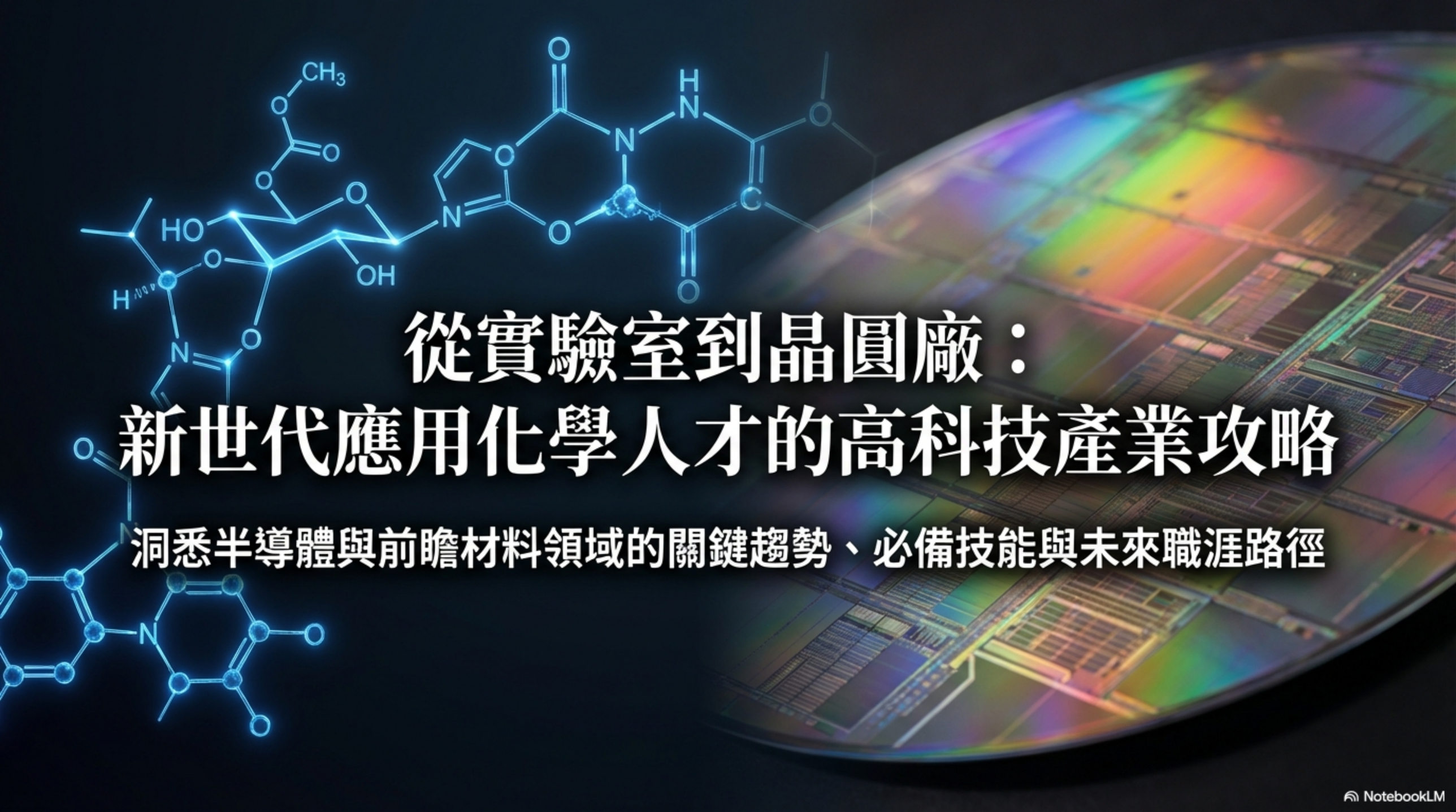


從實驗室到晶圓廠：新世代應用化學人才的高科技產業攻略

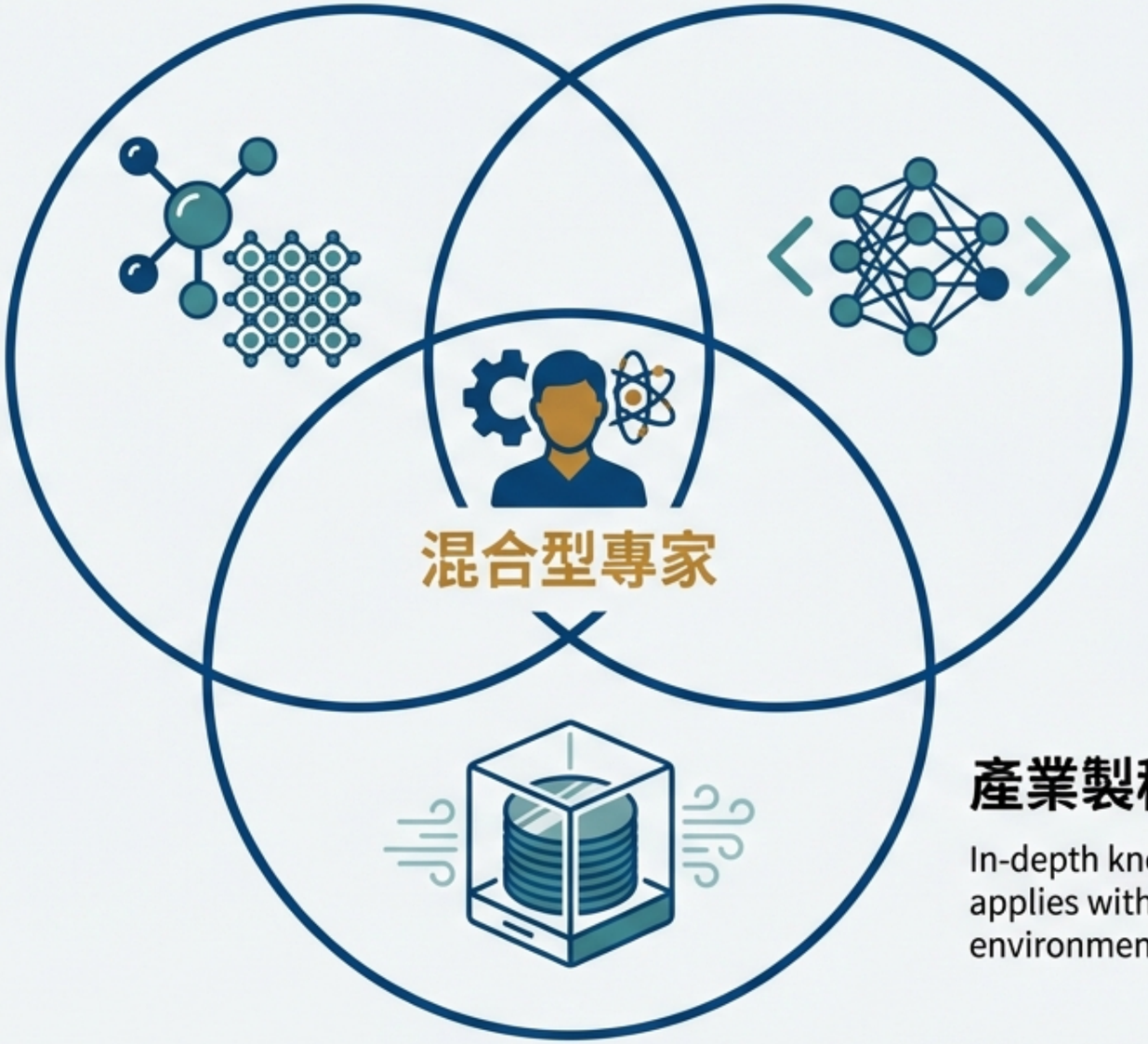
洞悉半導體與前瞻材料領域的關鍵趨勢、必備技能與未來職涯路徑

NotebookLM

未來的關鍵化學人才，是兼具分子科學、計算思維與跨域能力的「混合型專家」。

前瞻材料科學

Deep understanding of molecular principles and their application in semiconductor and nanotechnology contexts.



數據分析與AI

Proficiency in computational tools, simulation, and leveraging AI to accelerate R&D.

產業製程知識

In-depth knowledge of how chemistry applies within the high-tech manufacturing environment, especially in fabs.

NotebookLM

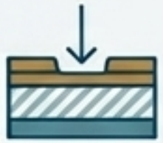
台灣的應用化學研究已深度對焦產業需求，鎖定先進半導體與奈米材料的技術缺口。



關鍵研究主軸 (Key Research Thrusts)

先進材料與奈米技術

Directly driven by the industry's push towards the 2奈米 process node.



光阻劑
(photoresists)



低介電常數材料
(low-k dielectrics)

生物醫藥與綠色能源

Strategic focus on small molecule drug design and sustainable energy materials.



小分子藥物設計



固態電池材料

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精通三大核心硬實力，是進入高科技產業的入場券。

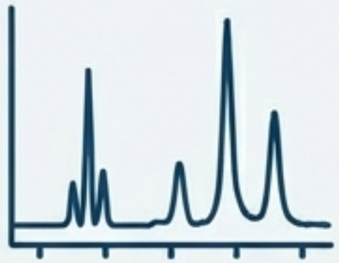


半導體製程化學

Deep understanding of micro-lithography (微影), etching (蝕刻), and Atomic Layer Deposition (ALD).

超微量分析與污染控制

Tracking impurities at the parts-per-trillion (兆分之一) level.



高階儀器分析

Mastery of interpreting data from core instruments for process diagnostics.

SEM/TEM, NMR, FTIR, Raman Spectroscopy



數據科學與模擬

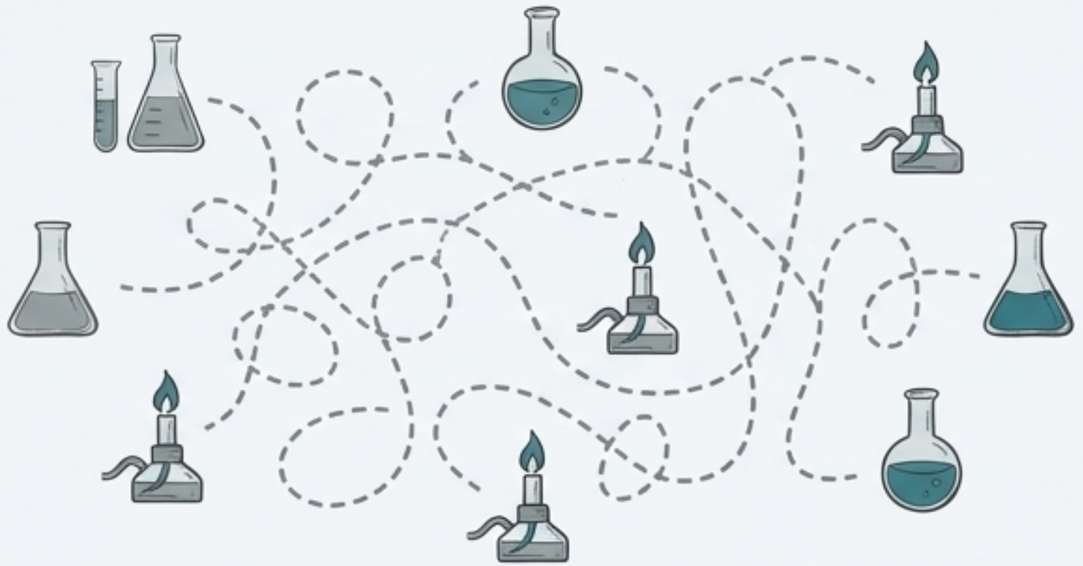
Familiarity with computational chemistry to predict material properties and reactions.

Density Functional Theory (DFT) and Molecular Dynamics (MD) simulations.

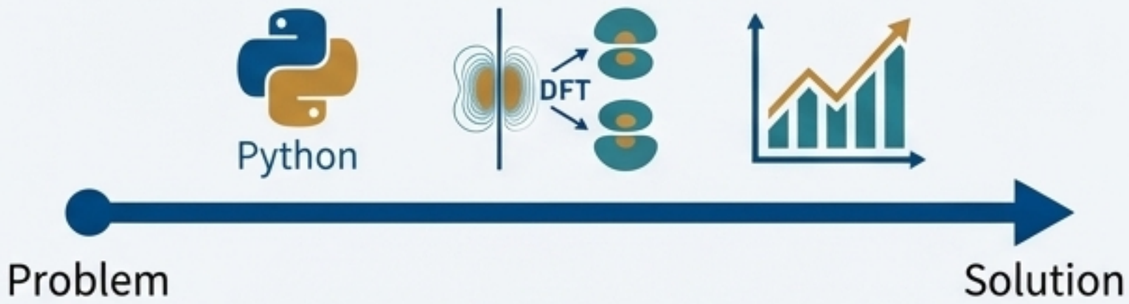
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從「試誤法」轉向「預測設計」，計算能力是研發效率的倍增器。

傳統「試誤法」 (Trial and Error)



現代「預測設計」 (Predictive Design)



關鍵計算工具 (Key Computational Tools)

程式設計：**Python**

Essential Libraries: **RDKit** (cheminformatics) and **DeepChem** (deep learning)

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卓越的邏輯、溝通與AI素養，是從優秀到卓越的關鍵。

根本原因分析 (RCA)

Using rigorous logic to diagnose problems, sifting through massive datasets to find the chemical source.



跨領域溝通

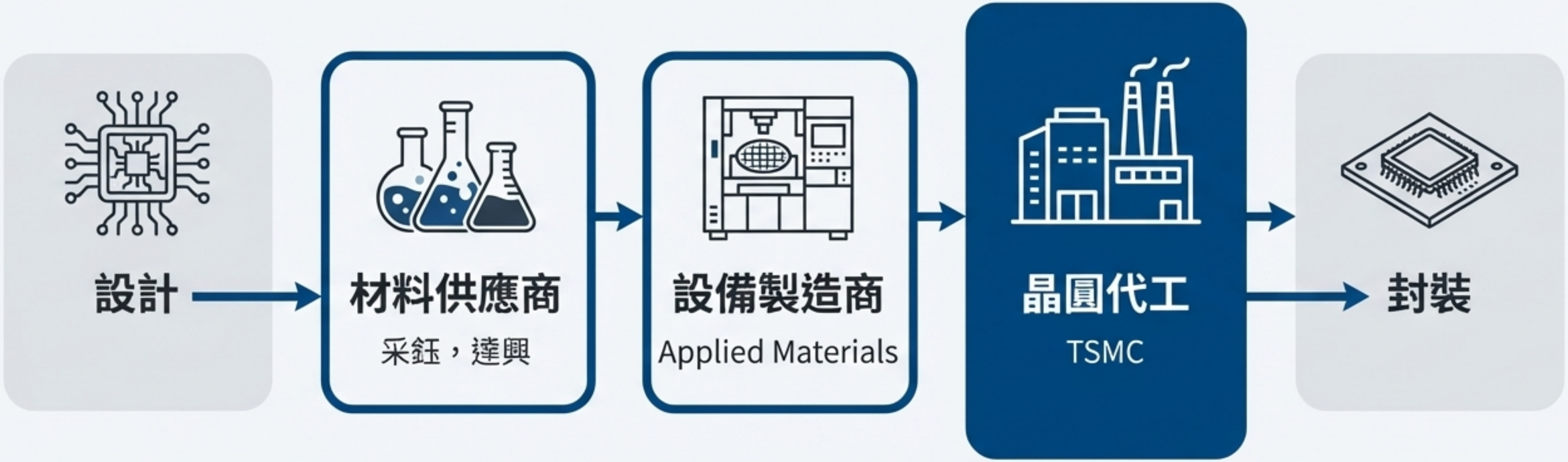
Translating complex chemical principles into language other engineers can understand and act on.

AI 就緒與批判性思維

An **AI就緒 (AI-Ready)** mindset, including **提示工程 (Prompt Engineering)** and critical validation of AI outputs.

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半導體產業鏈是應用化學人才的最大舞台，提供最多元的核心職位。



Key Opportunities for Chemists

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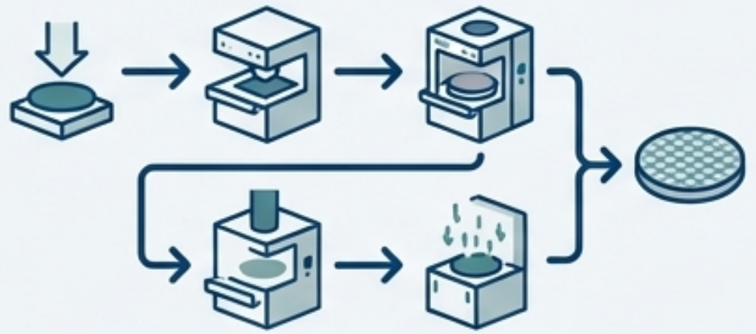
製程與整合工程師是晶圓廠的「指揮官」，運用化學知識確保產線穩定與良率。

製程工程師 (Process Engineer, PE)



- **焦虑**：負責一特定的模块（如、刻刻、光牙）。
- **核心任務**：調整化學參數（氣流、濃度配），維持線產程穩定性。
- **需要知識**：反應動率、雷斷 化學

製程整合工程師 (Process Integration Engineer, PIE)



- **焦虑**：監覽整是製造流程及解決最產品農業問題。
- **核心任務**：協諫不同模块，了解化學步驟的化交相互。
- **需要知識**：高上平化學逻輯、系統思考

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研發與良率工程師是「創新者」與「偵探」，負責開發新材料並揪出製程的化學污染源。

研發工程師 (R&D Engineer)
The 'Innovator'



- Developing new materials and processes for future technology nodes.
- Designing new photoresists, polymers, or nanomaterials.

良率提升工程師 (Yield Enhancement / QC Engineer)
The 'Detective'



- Monitoring defect density and using analytical chemistry to identify sources of contamination.
- Strong background in instrumental analysis and data mining.

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除了半導體，化學專業在生技醫藥與永續產業中同樣扮演不可或缺的角色。



製劑科學家 (Formulation Scientist)

Using polymer chemistry to design effective drug delivery systems.



藥物/分析化學家 (Medicinal/Analytical Chemist)

Designing new drug molecules or ensuring the purity of pharmaceutical products.



環安衛工程師 (EHS Engineer)

Managing chemical waste, risk assessment, and ensuring regulatory compliance with the rise of ESG.

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生成式AI正成為化學家的「新試管」，從根本上改變研究與實驗的工作模式。

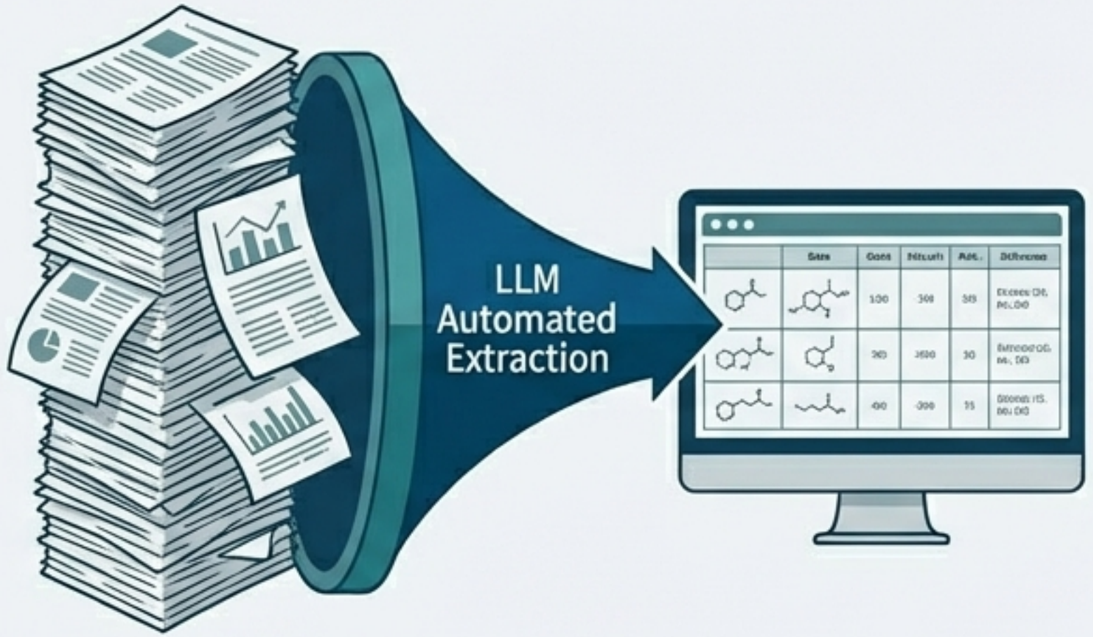


The chemist evolves from an operator of lab equipment to a conductor of digital tools.

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AI能將數月的文獻探勘縮短為數小時，並能預測最佳的化學合成路徑。

文獻探勘與知識提取



Saves months of manual reading and data compilation.

合成路徑規劃



Predicts reaction outcomes and recommends the most efficient synthesis pathways.

NotebookLM

AI實現了從「性質到分子」的逆向設計，並能自動解析複雜的光譜數據。

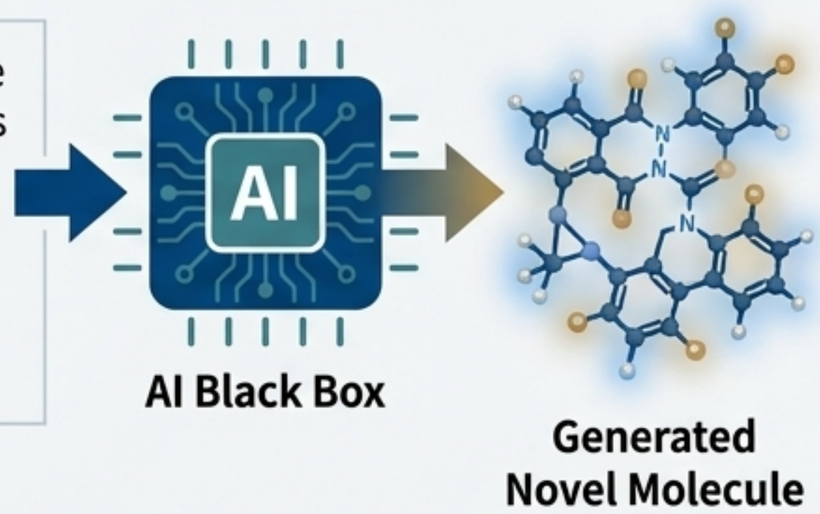
逆向分子設計
(Inverse Molecular Design)

Input Properties

Low-impedance for **2nm** process

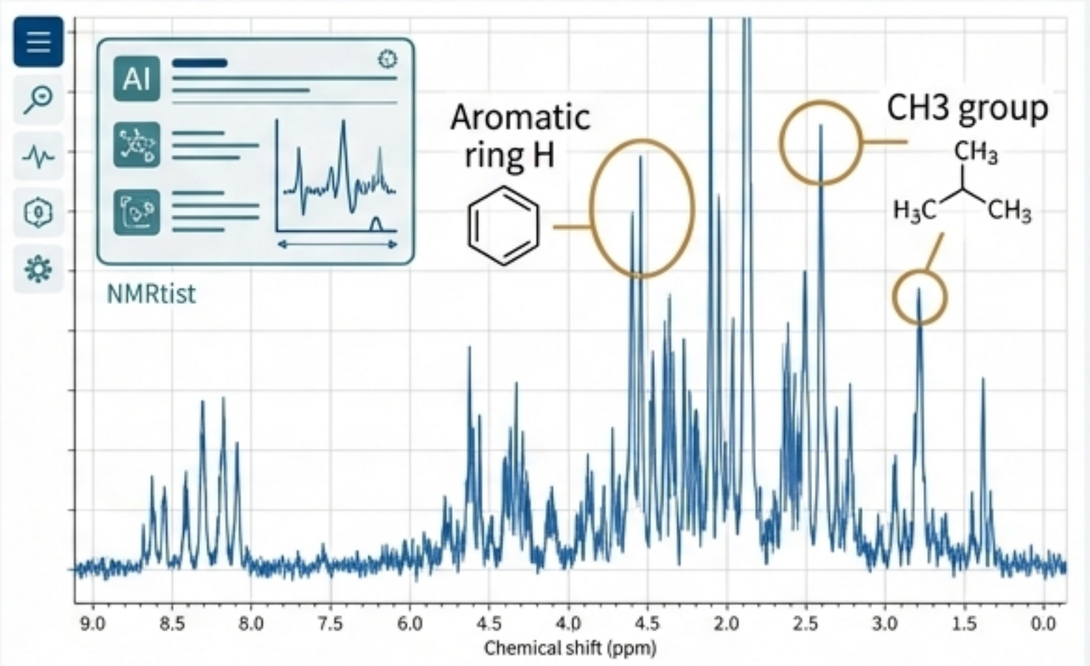
High thermal stability

Specific optical transparency



Moves from screening existing molecules to designing ideal ones from scratch.

光譜自動化解析
(Automated Spectral Analysis)



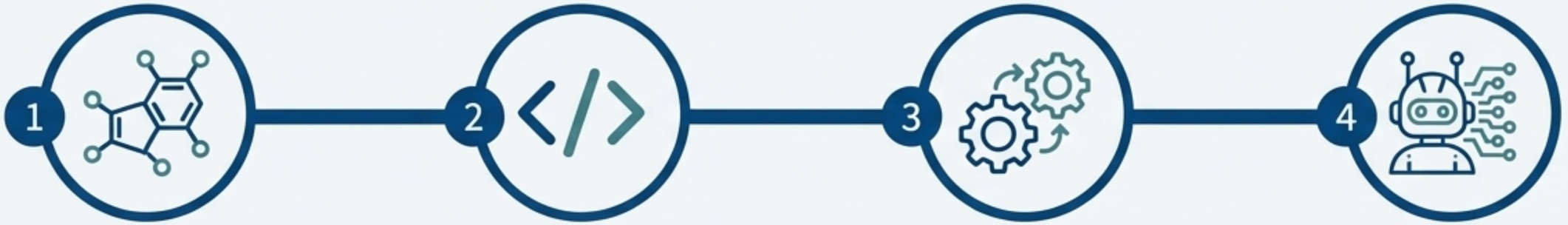
Reduces multi-week manual analysis to hours and increases accuracy (>83% for IR spectra).

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您的成功路徑，是將學術基礎、實務技能與數位工具融為一體。

鞏固核心化學

培養系統思維



鞏固核心化學

Master the fundamentals of organic, inorganic, physical, and analytical chemistry.

擁抱計算與數據

Actively learn Python, simulation tools (DFT/MD), and data analysis.

培養系統思維

Develop skills in RCA and cross-disciplinary communication to solve complex problems.

善用AI協作

Treat AI as a partner to accelerate research, innovate, and augment your expertise.

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未來，由您定義。
您不只是化學家，更是驅動台灣下一個科技世代的關鍵創新者。
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